



Lab Report:02

Course Code: ICT - 3104

Course Name: Computer Network Lab

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Objective

To design, configure, and evaluate a multi-subnet network topology comprising multiple PCs, switches, and routers, ensuring accurate IP addressing, proper connectivity, and efficient data transmission.

Theory

This experiment is grounded in the fundamentals of computer networking, including IP addressing, subnetting, and routing protocols. IP addresses are assigned to devices to facilitate communication within and across subnets. Subnetting divides a network into smaller, manageable segments, each with its own subnet mask (e.g., /24). Routers utilize routing tables to forward packets between subnets, while switches operate at the data link layer to manage intra-subnet traffic based on MAC addresses. The success of the topology depends on correct configuration and interoperability of these components.

Tools and Components

- ComputingDevices: 7 PCs (PC0 to PC6)
- NetworkingHardware:
 - i. 4 Cisco 2960-24TT Switches (Switch0 to Switch3)
 - ii. 3 Routers (Router0 to Router2)
- **Cabling:** Ethernet cables (CAT5e or higher)
- **Software:** Network configuration tools (e.g., Cisco IOS, terminal emulators)

Topology Overview

The network is structured into three distinct subnets interconnected via routers:

1. Subnet 1 (192.168.1.0/24):

- Devices: PC0 (192.168.1.2), PC1 (192.168.1.3)
- Switch: Switch0
- RouterInterface: Router0 (11.0.0.2)

2. Subnet 2 (192.168.2.0/24):

- Devices: PC2 (192.168.2.2), PC3 (192.168.2.3)
- Switch: Switch1
- RouterInterface: Router0 (12.0.0.2)

3. Subnet 3 (192.168.3.0/24):

- Devices: PC4 (192.168.3.2), PC5 (192.168.3.3), PC6 (192.168.3.4)
- Switches: Switch2, Switch3
- RouterInterfaces: Router1 (13.0.0.2), Router2 (13.0.0.3)

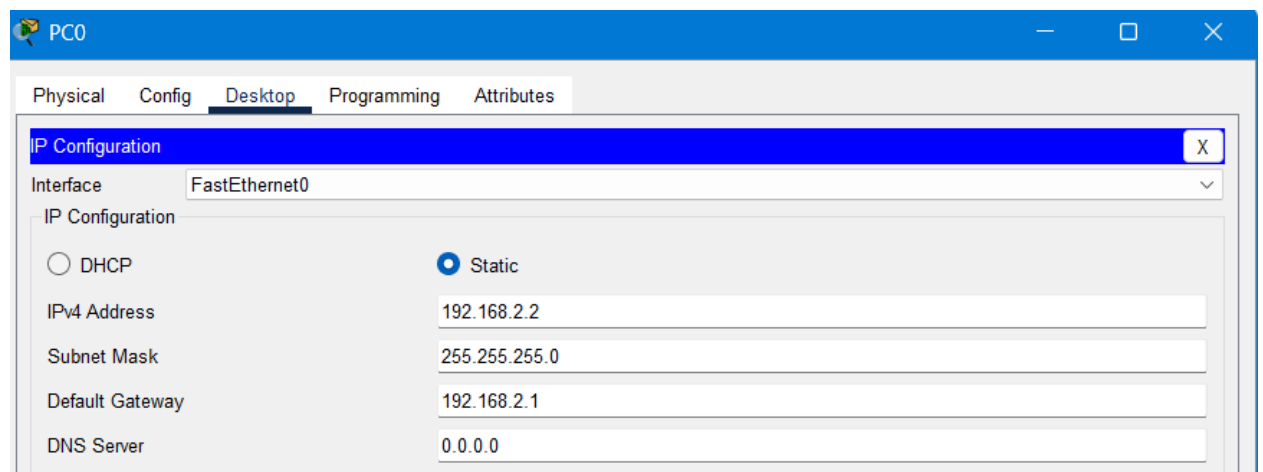
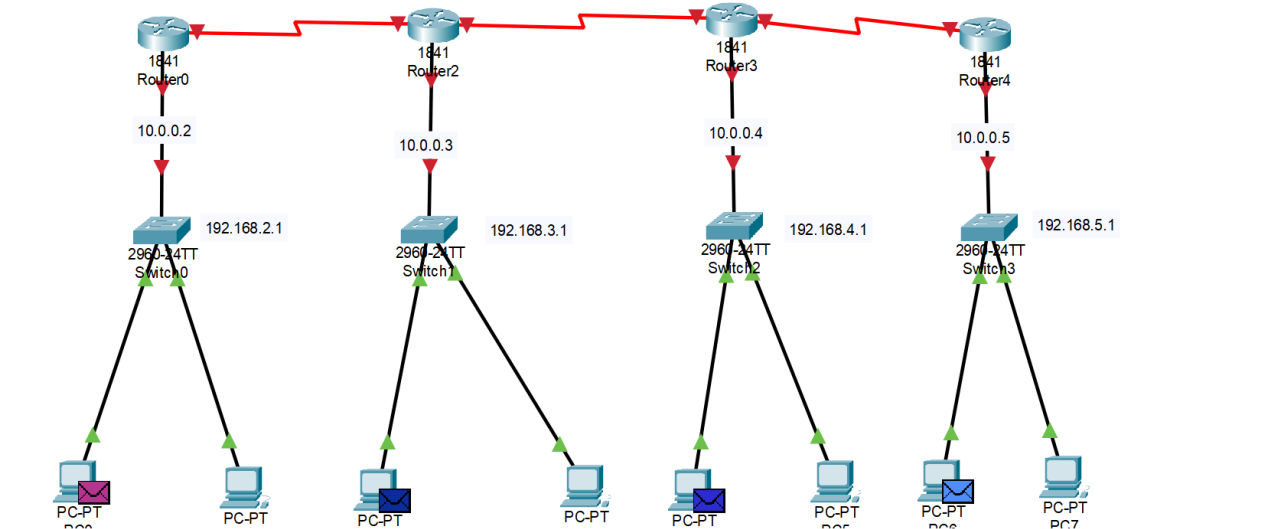
4. Inter-Router Connections:

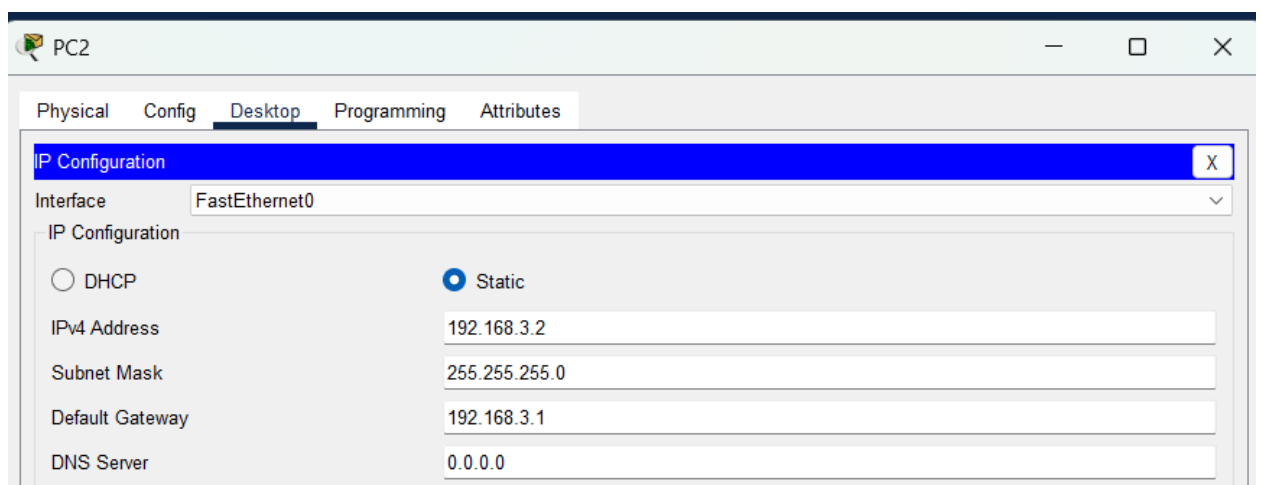
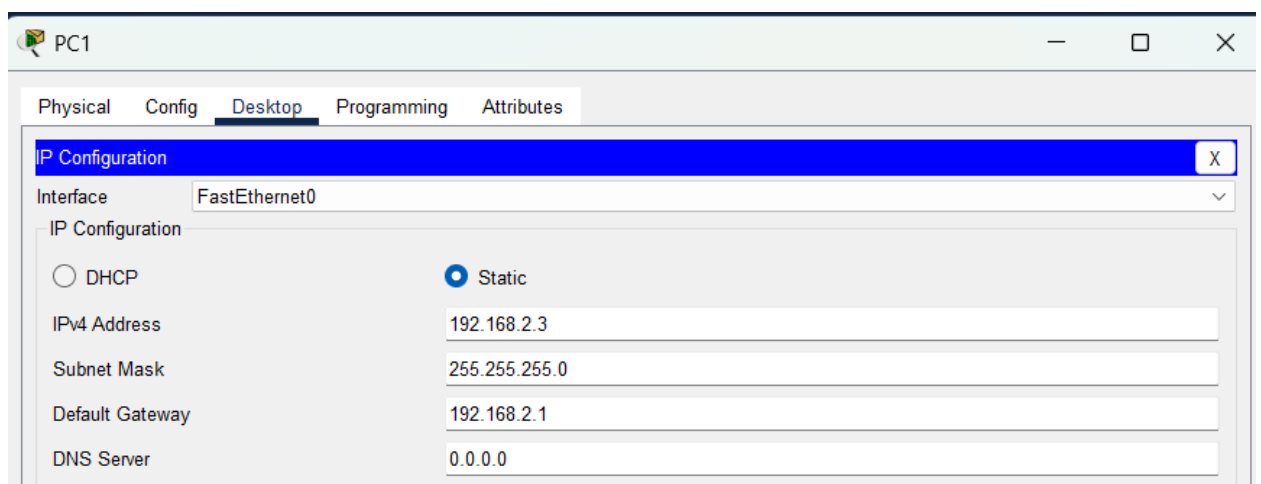
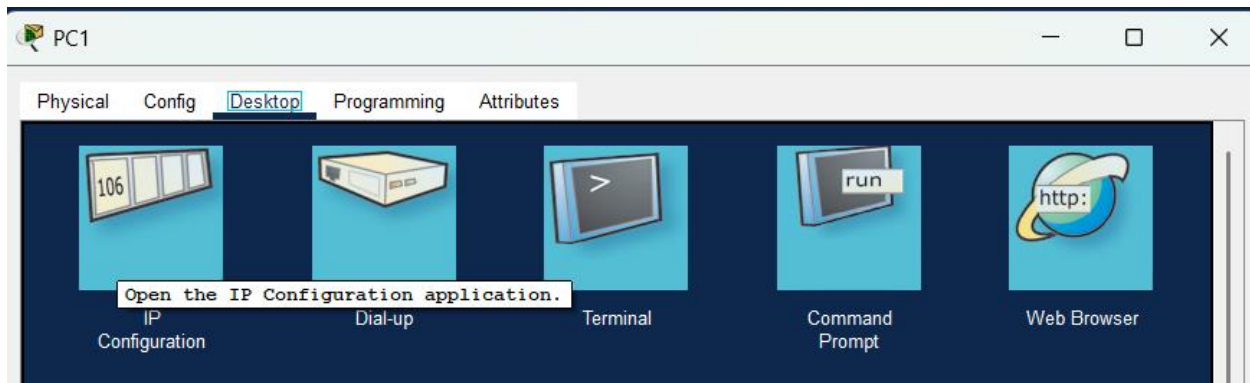
- Router0: 11.0.0.2 (Subnet 1), 12.0.0.2 (Subnet 2)
- Router1: 12.0.0.3 (Subnet 2), 13.0.0.2 (Subnet 3)
- Router2: 13.0.0.3 (Subnet 3)

Procedure

1. **Configuration:** Assign IP addresses to all PCs and router interfaces as per the topology diagram using appropriate network configuration tools.
2. **Physical Setup:** Connect devices using Ethernet cables according to the designated topology, ensuring secure and proper termination.
3. **Connectivity Testing:**
 - λ Perform ping tests within each subnet (e.g., PC0 to PC1) to verify local connectivity.
 - λ Conduct inter-subnet ping tests (e.g., PC0 to PC2) to confirm routing functionality.
4. **Documentation:** Record all IP configurations, connection statuses, and test results for analysis

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PC3

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.3.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.3.1

DNS Server 0.0.0.0

PC4

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.4.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.4.1

DNS Server 0.0.0.0

PC5

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.4.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.4.1

DNS Server 0.0.0.0

PC6

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.5.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.5.1

DNS Server 0.0.0.0

PC7

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

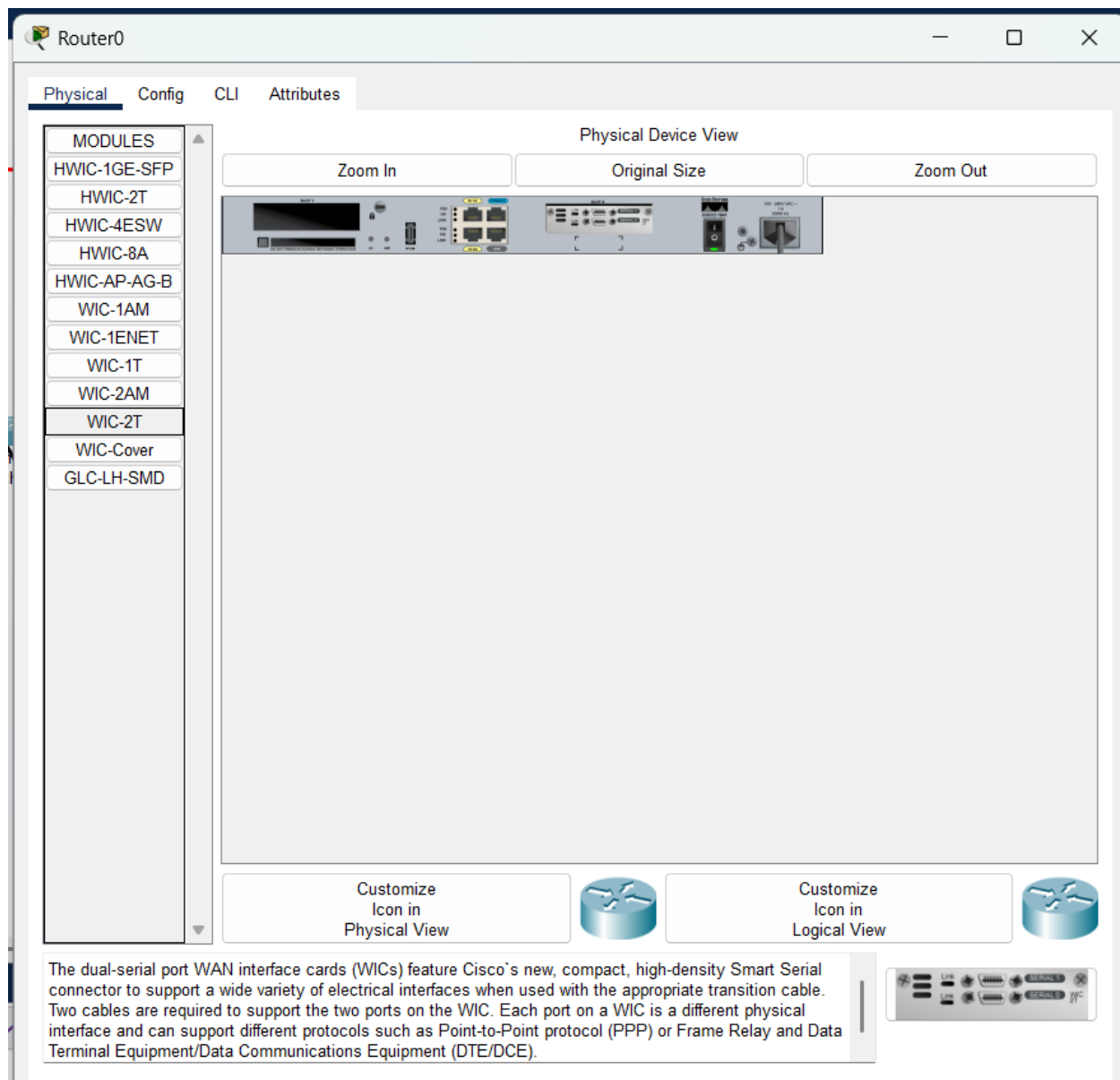
☐ DHCP ☒ Static

IPv4 Address 192.168.5.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.5.1

DNS Server 0.0.0.0



Result and Observation

- **Intra-Subnet Connectivity:** Ping tests within Subnet 1 (PC0 to PC1) and Subnet 2 (PC2 to PC3) yielded 100% success with response times averaging 1-2 ms.

- **Inter-Subnet Connectivity:** Ping tests between Subnet 1 and Subnet 2 (e.g., PC0 to PC2) were successful, with response times averaging 5-7 ms, indicating effective routing.
- **Subnet 3 Performance:** All devices (PC4 to PC6) communicated successfully within Subnet 3 and with Router1 and Router2, with no packet loss observed.
- **Network Stability:** No connectivity issues or packet drops were recorded during a 15-minute observation period.

Conclusion

The multi-subnet network topology was successfully designed and implemented, demonstrating accurate IP addressing, robust intra- and inter-subnet connectivity, and reliable routing. The configuration met all objectives, with consistent performance across all tested scenarios. For enhanced scalability and resilience, future implementations should consider VLAN segmentation and redundant routing paths.